SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
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॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java
ASSIGNMENT No : 7
TITLE : Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

EXERCISE

1.Create a **Shape application** where Circle and Rectangle inherit from Shape and calculate area.

2. Develop a **E-Commerce Product System** where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

CODE

Main.java ./ | .gitattributes | Student.java U | bank.java U | book.java U

exp7 > Main.java > Shape > area()

```java
import java.util.Scanner;

// Parent class
class Shape {
    void area() {
    }
}

// Child class Circle
class Circle extends Shape {
    double radius;

    Circle(double radius) {
        this.radius = radius;
    }

    void area() {
        double area = 3.14 * radius * radius;
        System.out.println("Area of Circle = " + area);
    }
}

// Child class Rectangle
class Rectangle extends Shape {
    double length, breadth;

    Rectangle(double length, double breadth) {
        this.length = length;
        this.breadth = breadth;
    }

    void area() {
        double area = length * breadth;
        System.out.println("Area of Rectangle = " + area);
    }
}

```

```java

public class Main {
    Run main | Debug main
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


        System.out.print("Enter Radius of Circle: ");
        double radius = sc.nextDouble();

        Circle c = new Circle(radius);
        c.area();

        System.out.print("\nEnter Length of Rectangle: ");
        double length = sc.nextDouble();

        System.out.print("Enter Breadth of Rectangle: ");
        double breadth = sc.nextDouble();

        Rectangle r = new Rectangle(length, breadth);
        r.area();

        sc.close();
    }
}
```


```java

public class Main {
    Run main | Debug main
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);


        System.out.print("Enter Radius of Circle: ");
        double radius = sc.nextDouble();

        Circle c = new Circle(radius);
        c.area();

        System.out.print("\nEnter Length of Rectangle: ");
        double length = sc.nextDouble();

        System.out.print("Enter Breadth of Rectangle: ");
        double breadth = sc.nextDouble();

        Rectangle r = new Rectangle(length, breadth);
        r.area();

        sc.close();
    }
}
```

OUTPUT

```
1 error
● raghav_oze@raghavs-MacBook-Air-3 exp8 % cd
● raghav_oze@raghavs-MacBook-Air-3 ~ % cd ~/Desktop/java/exp7
● raghav_oze@raghavs-MacBook-Air-3 exp7 % javac Main.java
● raghav_oze@raghavs-MacBook-Air-3 exp7 % java Main.java
Enter Radius of Circle: 4
Area of Circle = 50.24

Enter Length of Rectangle: 4
Enter Breadth of Rectangle: 4
Area of Rectangle = 16.0
raghav_oze@raghavs-MacBook-Air-3 exp7 %
```

```
● raghav_oze@raghavs-MacBook-Air-3 exp7 % clear
● raghav_oze@raghavs-MacBook-Air-3 exp7 % java mains.java
Enter Electronic Product Details
Product ID: 001
Product Name: macbook
Price: 80000

Enter Clothing Product Details
Product ID: 002
Product Name: raincoat
Price: 2000

Enter Grocery Product Details
Product ID: 003
Product Name: dark choclate
Price: 130

Electronic Product
Product ID   : 1
Product Name : macbook
Price        : ₹80000.0

Clothing Product
Product ID   : 2
Product Name : raincoat
Price        : ₹2000.0

Grocery Product
Product ID   : 3
Product Name : dark choclate
Price        : ₹130.0
raghav_oze@raghavs-MacBook-Air-3 exp7 %
```

CODE

exp7 > J hfdc

```java
import java.util.Scanner;

interface Product {
    void displayDetails();
}

// Parent Class
class ProductDetails {
    int productId;
    String productName;
    double price;

    ProductDetails(int productId, String productName, double price) {
        this.productId = productId;
        this.productName = productName;
        this.price = price;
    }
}


class Electronic extends ProductDetails implements Product {

    Electronic(int productId, String productName, double price) {
        super(productId, productName, price);
    }

    public void displayDetails() {
        System.out.println("\nElectronic Product");
        System.out.println("Product ID   : " + productId);
        System.out.println("Product Name : " + productName);
        System.out.println("Price        : ₹" + price);
    }
}
```

```java

class Clothing extends ProductDetails implements Product {

    Clothing(int productId, String productName, double price) {
        super(productId, productName, price);
    }

    public void displayDetails() {
        System.out.println("\nClothing Product");
        System.out.println("Product ID   : " + productId);
        System.out.println("Product Name : " + productName);
        System.out.println("Price        : ₹" + price);
    }
}


class Grocery extends ProductDetails implements Product {

    Grocery(int productId, String productName, double price) {
        super(productId, productName, price);
    }

    public void displayDetails() {
        System.out.println("\nGrocery Product");
        System.out.println("Product ID   : " + productId);
        System.out.println("Product Name : " + productName);
        System.out.println("Price        : ₹" + price);
    }
}


public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

```

```java
        System.out.println("Enter Electronic Product Details");
        System.out.print("Product ID: ");
        int id1 = sc.nextInt();
        sc.nextLine();

        System.out.print("Product Name: ");
        String name1 = sc.nextLine();

        System.out.print("Price: ");
        double price1 = sc.nextDouble();
        sc.nextLine();

        Electronic e = new Electronic(id1, name1, price1);



        System.out.println("\nEnter Clothing Product Details");
        System.out.print("Product ID: ");
        int id2 = sc.nextInt();
        sc.nextLine();

        System.out.print("Product Name: ");
        String name2 = sc.nextLine();

        System.out.print("Price: ");
        double price2 = sc.nextDouble();
        sc.nextLine();

        Clothing c = new Clothing(id2, name2, price2);
```

```
        System.out.println("\nEnter Grocery Product Details");
        System.out.print("Product ID: ");
        int id3 = sc.nextInt();
        sc.nextLine();

        System.out.print("Product Name: ");
        String name3 = sc.nextLine();

        System.out.print("Price: ");
        double price3 = sc.nextDouble();

        Grocery g = new Grocery(id3, name3, price3);

        e.displayDetails();
        c.displayDetails();
        g.displayDetails();

    }
}
```

CONCLUSION

We have successfully learnt and implemented the use of inheritance in two programs